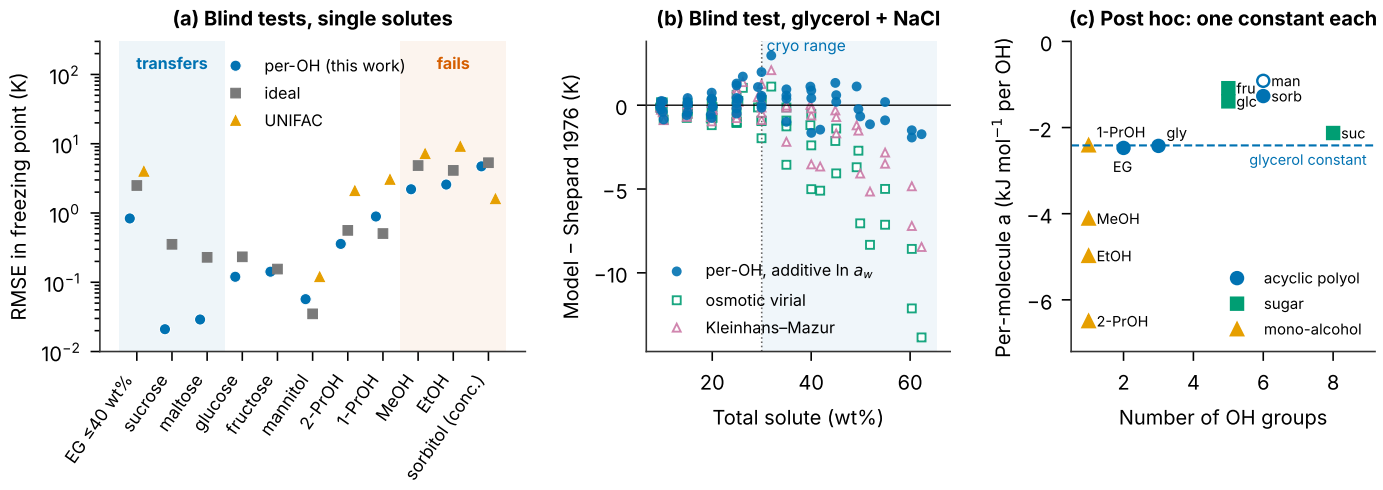
(a) Blind tests, single solutes
transfers
fails
per-OH (this work)
ideal
UNIFAC
RMSE in freezing point (K)
10^2
10^1
10^0
10^{-1}
10^{-2}
EG ≤40 wt%
sucrose
maltose
glucose
fructose
mannitol
2-PrOH
1-PrOH
MeOH
EtOH
sorbitol (conc.)
(b) Blind test, glycerol + NaCl
cryo range
per-OH, additive ln a_w
osmotic virial
Kleinhans–Mazur
Model − Shepard 1976 (K)
0
−5
−10
20
40
60
Total solute (wt%)
(c) Post hoc: one constant each
Per-molecule a (kJ mol^{-1} per OH)
0
−2
−4
−6
fru
glc
man
sorb
1-PrOH
gly
EG
suc
glycerol constant
MeOH
EtOH
2-PrOH
acyclic polyol
sugar
mono-alcohol
2
4
6
8
Number of OH groups